1) SQL to create a Customer2 table using appropriate data types that has the following columns: CustomerID, First Name, Last Name, Date of Birth, Address, City, State, Zip Code. CustomerID must be defined as an identity field that starts with 100.

```sql
Use Meijer
CREATE TABLE [dbo].[Customer2]( --creates table
    [CustomerID] [int] IDENTITY(100,1) NOT NULL, --sets identity
    [FName] [nvarchar](50) NOT NULL,
    [LName] [nvarchar](50) NOT NULL,
    [Address] [nvarchar](100) NULL,
    [DateOfBirth] [date] NULL,
    [City] [nvarchar](50) NULL,
    [State] [nvarchar](50) NULL,
    [ZipCode] [nvarchar](10) NULL,
 CONSTRAINT [PK_Customer2] PRIMARY KEY CLUSTERED --set primary key
(
    [CustomerID] ASC --ascending order
)WITH (PAD_INDEX = OFF, STATISTICS_NORECOMPUTE = OFF,
IGNORE_DUP_KEY = ON, ALLOW_ROW_LOCKS = ON,
ALLOW_PAGE_LOCKS = ON, OPTIMIZE_FOR_SEQUENTIAL_KEY = OFF) ON [PRIMARY]
) ON [PRIMARY]
GO
CREATE INDEX FNameIDX  --creates index
On Customer2 (FName);
```

105 % 21 0

Messages

```
Commands completed successfully.

Completion time: 2026-03-29T15:18:01.6816781-04:00
```

2-IN-1\SQLE...o.Customer2 | SQL3 notes.sq...-1\wills (69))*

Column Name	Data Type	Allow Nulls
CustomerID	int	☐
FName	nvarchar(50)	☐
LName	nvarchar(50)	☐
Address	nvarchar(100)	☑
DateOfBirth	date	☑
City	nvarchar(50)	☑
State	nvarchar(50)	☑
ZipCode	nvarchar(10)	☑
		☐

2) SQL to define the primary key in the CustomerID column. (SQL must not be generated)

```sql
Use Meijer
CREATE TABLE [dbo].[Customer2]( --creates table
    [CustomerID] [int] IDENTITY(100,1) NOT NULL, --sets identity
    [FName] [nvarchar](50) NOT NULL,
    [LName] [nvarchar](50) NOT NULL,
    [Address] [nvarchar](100) NULL,
    [DateOfBirth] [date] NULL,
    [City] [nvarchar](50) NULL,
    [State] [nvarchar](50) NULL,
    [ZipCode] [nvarchar](10) NULL,
 CONSTRAINT [PK_Customer2] PRIMARY KEY CLUSTERED --set primary key
(
    [CustomerID] ASC --ascending order
)WITH (PAD_INDEX = OFF, STATISTICS_NORECOMPUTE = OFF,
IGNORE_DUP_KEY = ON, ALLOW_ROW_LOCKS = ON,
ALLOW_PAGE_LOCKS = ON, OPTIMIZE_FOR_SEQUENTIAL_KEY = OFF) ON [PRIMARY]
) ON [PRIMARY]
```

105 % 21 0

Messages

```
Commands completed successfully.

Completion time: 2026-03-29T15:53:34.4125045-04:00
```

3) SQL to define an index using the primary key and a second index using Last Name and First Name. (SQL must not be generated)

2-IN-1\SQLE...bo.Customer2 | SQL3 notes.s...1\wills (69))*

```sql
CREATE INDEX NameIDX  --creates index
On Customer2 (FName, LName);
```

105 % 21 0

Messages

```
Commands completed successfully.

Completion time: 2026-03-29T15:53:34.4125045-04:00
```

4) SQL to insert fifty rows of unique data into the Customer2 table.

```sql
USE [Meijer]
GO

INSERT INTO [dbo].[Customer2] (FName, LName, Address, DateOfBirth, City, State, ZipCode) VALUES
('James', 'Holloway', '142 Birchwood Dr', '1985-03-12', 'Grand Rapids', 'Michigan', '49501'),
('Maria', 'Gutierrez', '87 Lakeview Ave', '1992-07-24', 'Lansing', 'Michigan', '48901'),
('Derek', 'Okafor', '319 Maple St', '1978-11-05', 'Detroit', 'Michigan', '48201'),
('Sandra', 'Petersen', '654 Elm Ct', '1990-01-30', 'Ann Arbor', 'Michigan', '48103'),
('Tyler', 'Nguyen', '22 Pinehurst Blvd', '1988-06-17', 'Kalamazoo', 'Michigan', '49001'),
('Rachel', 'Simmons', '501 Oak Ridge Rd', '1995-09-08', 'Flint', 'Michigan', '48501'),
('Kevin', 'Barnes', '78 Sunset Ln', '1982-04-22', 'Saginaw', 'Michigan', '48601'),
('Brittany', 'Holt', '213 Willowbrook Way', '1999-12-01', 'Traverse City', 'Michigan', '49684'),
('Marcus', 'Fitzgerald', '990 Cedar Ave', '1975-08-14', 'Marquette', 'Michigan', '49855'),
('Lauren', 'Yamamoto', '45 Riverside Dr', '1993-02-28', 'Midland', 'Michigan', '48640'),
('Nathan', 'Kowalski', '118 Aspen Ct', '1986-10-19', 'Bay City', 'Michigan', '48706'),
('Amber', 'Richardson', '367 Foxglove Ln', '1997-05-03', 'Holland', 'Michigan', '49423'),
('Justin', 'Morales', '829 Highview Terr', '1980-07-11', 'Muskegon', 'Michigan', '49440'),
('Heather', 'Chambers', '56 Briarwood Dr', '1991-03-25', 'Jackson', 'Michigan', '49201'),
('Andre', 'Washington', '742 Northfield Rd', '1984-09-16', 'Pontiac', 'Michigan', '48341'),
('Cynthia', 'Larsen', '303 Spruce Hill Ave', '1976-12-09', 'Dearborn', 'Michigan', '48120'),
('Brandon', 'Patel', '11 Meadowview Ln', '1994-04-07', 'Warren', 'Michigan', '48088'),
('Stephanie', 'OBrien', '488 Walnut St', '1989-08-30', 'Sterling Heights', 'Michigan', '48310'),
('Christopher', 'Evans', '675 Linden Blvd', '1983-01-14', 'Livonia', 'Michigan', '48150'),
('Ashley', 'Thompson', '234 Copper Ridge Rd', '1998-06-22', 'Royal Oak', 'Michigan', '48067'),
('Darnell', 'Jefferson', '109 Ironwood Dr', '1971-11-02', 'Southfield', 'Michigan', '48075'),
('Megan', 'Sullivan', '560 Cloverleaf Ct', '1996-03-18', 'Troy', 'Michigan', '48083'),
('Jonathan', 'Kim', '814 Overlook Pass', '1987-07-29', 'Auburn Hills', 'Michigan', '48326'),
('Vanessa', 'Cruz', '27 Ferndale Ave', '1993-10-10', 'Ferndale', 'Michigan', '48220'),
('Patrick', 'Murray', '391 Stone Gate Dr', '1979-02-05', 'Portage', 'Michigan', '49024'),
```

105 % No issues found

Messages

```
(50 rows affected)

Completion time: 2026-03-29T16:05:12.4877566-04:00
```

	CustomerID	FName	LName	Address	DateOfBirth	City	State	ZipCode
1	100	James	Holloway	142 Birchwood Dr	1985-03-12	Grand Rapids	Michigan	49501
2	101	Maria	Gutierrez	87 Lakeview Ave	1992-07-24	Lansing	Michigan	48901
3	102	Derek	Okafor	319 Maple St	1978-11-05	Detroit	Michigan	48201
4	103	Sandra	Petersen	654 Elm Ct	1990-01-30	Ann Arbor	Michigan	48103
5	104	Tyler	Nguyen	22 Pinehurst Blvd	1988-06-17	Kalamazoo	Michigan	49001
6	105	Rachel	Simmons	501 Oak Ridge Rd	1995-09-08	Flint	Michigan	48501
7	106	Kevin	Barnes	78 Sunset Ln	1982-04-22	Saginaw	Michigan	48601
8	107	Brittany	Holt	213 Willowbrook Way	1999-12-01	Traverse City	Michigan	49684
9	108	Marcus	Fitzgerald	990 Cedar Ave	1975-08-14	Marquette	Michigan	49855
10	109	Lauren	Yamamoto	45 Riverside Dr	1993-02-28	Midland	Michigan	48640
11	110	Nathan	Kowalski	118 Aspen Ct	1986-10-19	Bay City	Michigan	48706
12	111	Amber	Richards...	367 Foxglove Ln	1997-05-03	Holland	Michigan	49423
13	112	Justin	Morales	829 Highview Terr	1980-07-11	Muskegon	Michigan	49440
14	113	Heat...	Chambers	56 Briarwood Dr	1991-03-25	Jackson	Michigan	49201
15	114	Andre	Washing...	742 Northfield Rd	1984-09-16	Pontiac	Michigan	48341
16	115	Cynthia	Larsen	303 Spruce Hill Ave	1976-12-09	Dearborn	Michigan	48120
17	116	Bran...	Patel	11 Meadowview Ln	1994-04-07	Warren	Michigan	48088
18	117	Step...	OBrien	488 Walnut St	1989-08-30	Sterling Hei...	Michigan	48310
19	118	Christ...	Evans	675 Linden Blvd	1983-01-14	Livonia	Michigan	48150
20	119	Ashley	Thompson	234 Copper Ridge Rd	1998-06-22	Royal Oak	Michigan	48067
21	120	Darnell	Jefferson	109 Ironwood Dr	1971-11-02	Southfield	Michigan	48075
22	121	Megan	Sullivan	560 Cloverleaf Ct	1996-03-18	Troy	Michigan	48083
23	122	Jonat...	Kim	814 Overlook Pass	1987-07-29	Auburn Hills	Michigan	48326
24	123	Vane...	Cruz	27 Ferndale Ave	1993-10-10	Ferndale	Michigan	48220
25	124	Patrick	Murray	391 Stone Gate Dr	1979-02-05	Portage	Michigan	49024
26	125	Dani...	Webb	143 Creekside Blvd	1990-05-27	Okemos	Michigan	48864
27	126	Greg...	Scott	720 Blue Heron Way	1985-08-13	East Lansing	Michigan	48823

5) SQL to truncate and drop the Customer2 table.

```sql
use Meijer
go
Select * from Customer2 --show table with records

Truncate table Customer2  --deleted records
Select * from Customer2

Drop table Customer2  --deletes table
Select * from Customer2
```

21 0

sults Messages

```
(50 rows affected)

(0 rows affected)
Msg 208, Level 16, State 1, Line 73
Invalid object name 'Customer2'.

Completion time: 2026-03-29T16:08:28.1902398-04:00
```

6) SQL to create a view of the Customer2 table using all of the columns in the table except Date of Birth.

```sql
create view vw_Customer2 as --creates view
select CustomerID, FName, LName, Address, City, State, ZipCode  --gathers this information
from Customer2
```

24 0

ssages

```
Commands completed successfully.

Completion time: 2026-03-29T16:14:31.5461711-04:00
```

7) SQL that uses the view to display all Customers in the Customer2 table sorted by First Name in descending order.

```sql
select * from vw_Customer2
order by FName desc
```

105 % 23 0

Results Messages

	CustomerID	FName	LName	Address	City	State	ZipCode
1	142	Zachary	Campbell	221 Riverbend Dr	Port Huron	Michigan	48060
2	140	Wesley	Parker	316 Deerfield Ave	Adrian	Michigan	49221
3	123	Vanessa	Cruz	27 Ferndale Ave	Ferndale	Michigan	48220
4	104	Tyler	Nguyen	22 Pinehurst Blvd	Kalamazoo	Michigan	49001
5	148	Trevor	Long	298 Morning Glory Dr	Alma	Michigan	48801
6	134	Timothy	Gray	95 Woodland Ave	Houghton	Michigan	49931
7	127	Tiffany	Allen	58 Ridgecrest Dr	Ypsilanti	Michigan	48197
8	145	Tanya	Cooper	430 Greenfield Way	Ionia	Michigan	48846
9	117	Stephanie	OBrien	488 Walnut St	Sterling Heights	Michigan	48310
10	136	Shawn	Coleman	748 Clearwater Blvd	Gaylord	Michigan	49735
11	103	Sandra	Petersen	654 Elm Ct	Ann Arbor	Michigan	48103
12	128	Samuel	Robinson	476 Pinewood Ct	Mount Pleasant	Michigan	48858
13	132	Ryan	Hughes	642 Harborview Dr	Escanaba	Michigan	49829
14	105	Rachel	Simmons	501 Oak Ridge Rd	Flint	Michigan	48501
15	146	Phillip	Ward	179 Stonegate Blvd	Charlotte	Michigan	48813
16	124	Patrick	Murray	391 Stone Gate Dr	Portage	Michigan	49024
17	129	Nicole	Foster	209 Brookhaven Rd	Cadillac	Michigan	49601
18	110	Nathan	Kowalski	118 Aspen Ct	Bay City	Michigan	48706
19	131	Monica	Sanders	17 Sunridge Ln	Petoskey	Michigan	49770
20	121	Megan	Sullivan	560 Cloverleaf Ct	Troy	Michigan	48083
21	101	Maria	Gutierrez	87 Lakeview Ave	Lansing	Michigan	48901
22	108	Marcus	Fitzgerald	990 Cedar Ave	Marquette	Michigan	49855
23	109	Lauren	Yamam...	45 Riverside Dr	Midland	Michigan	48640
24	141	Latoya	Phillips	584 Crestwood Ct	Monroe	Michigan	48161
25	106	Kevin	Barnes	78 Sunset Ln	Saginaw	Michigan	48601
26	144	Keith	Bailey	65 Lakewood Shor...	Clare	Michigan	48617
27	112	Justin	Morales	829 Highview Terr	Muskegon	Michigan	49440
28	122	Jonathan	Kim	814 Overlook Pass	Auburn Hills	Michigan	48326

8) One sql routine with at least 3 If statements and confirmation that they work.

```
Declare @Counter int = 3
If @Counter < 2
    Print 'Less than 2'
If @Counter = 2
    Print 'equal 2'
If @Counter > 2
    Print 'Greater than 2 | The value is: ' + CAST(@Counter AS VARCHAR(10))
```

87 % 22 0

Messages

```
Greater than 2 | The value is: 3

Completion time: 2026-03-26T09:45:26.6534432-04:00
```

9) One sql routine with If, Else-If and confirmation that they work.

```
Declare @Counter int = 3
If @Counter < 2
    Print 'Less than 2'
Else If @Counter = 2
    Print 'equal 2'
Else
    Print 'Greater than 2 | The value is: ' + CAST(@Counter AS VARCHAR(10))
```

96 % 22 0 Ln: 93, Ch: 73, Col: 76

Messages

```
Greater than 2 | The value is: 3

Completion time: 2026-03-26T09:46:13.4151122-04:00
```

10) One Case statement with at least 3 When options and confirmation that they work.

```
Declare @Counter int = 1
PRINT
    CASE
        WHEN @Counter < 2 THEN 'Less than 2'
        WHEN @Counter = 2 THEN 'equal 2'
        ELSE CONCAT ('Greater than 2 | value is: ', @Counter)
    END;
```

87 % 22 0

Messages

```
Less than 2

Completion time: 2026-03-26T09:43:50.9565091-04:00
```

11) Alter table sql to add an integer column to an existing table that allows nulls.

```
ALTER TABLE Customer
ADD Age tinyint;
```

87 % 22 0

Messages

```
Commands completed successfully.

Completion time: 2026-03-26T09:42:38.4178995-04:00
```

12) SQL update routine with error checking and error logging. Confirmation statement that generates logged error.

```sql
Use Meijer
Declare @ErrorHold int = 0
INSERT INTO Customer
    (CustomerID, FName)
Values
    (1, 'John')
set @ErrorHold = @@error
print CONCAT('@@error is: ', @ErrorHold)

If @ErrorHold != 0
    Insert into ErrorAudit
        (ErrorNbr)
    Values
        (@ErrorHold)
```

7 % 22 0

Messages

```
Msg 544, Level 16, State 1, Line 125
Cannot insert explicit value for identity column in table 'Customer' when IDENTITY_INSERT is set to OFF.
@@error is: 544

(1 row affected)

Completion time: 2026-03-26T09:38:59.7837500-04:00
```

13) SQL statement that defines and uses variables.

```sql
Use Meijer
Declare @ErrorHold int = 0 --declaring the variable as an int and making it equal to 0
INSERT INTO Customer
    (CustomerID, FName)
Values
    (1, 'John')
set @ErrorHold = @@error --setting the variable to the @@error number
print CONCAT('@@error is: ', @ErrorHold)

--when the @ErrorHold variable is no longer zero it inserts the number into the ErrorAudi table
If @ErrorHold != 0
    Insert into ErrorAudit
        (ErrorNbr)
    Values
        (@ErrorHold)
```

No issues found Ln: 10, Ch: 96 TABS

ssages

```
Msg 544, Level 16, State 1, Line 3
Cannot insert explicit value for identity column in table 'Customer' when IDENTITY_INSERT is set to OFF.
@@error is: 544

(1 row affected)

Completion time: 2026-03-31T11:26:27.0056213-04:00
```

14) Store procedure that accepts a variable and has error checking and logging as part of the execute stored procedure statement.

```
--Creates a stored procedure
create proc sp_Insert_ErrorAudit @ErrorHold int as
INSERT INTO ErrorAudit
    (ErrorNbr)
Values
    (@ErrorHold)
```

2 0

ssages

```
Commands completed successfully.

Completion time: 2026-04-02T21:46:54.8216215-04:00
```

15) SQL insert statement that inserts ten rows of data into a new table named Cars that includes at least 5 columns and the data varies for each record by using variables that are incremented and/or changed within the routine for each record.

```sql
-- Create table
CREATE TABLE Cars (
    CarID INT PRIMARY KEY,
    Make NVARCHAR(50),
    Model NVARCHAR(50),
    Year INT,
    Price DECIMAL(10,2),
    Mileage INT
);

-- Declare variables
DECLARE @CarID INT = 1;
DECLARE @Make NVARCHAR(50);
DECLARE @Model NVARCHAR(50);
DECLARE @Year INT = 2015;
DECLARE @Price DECIMAL(10,2) = 15000;
DECLARE @Mileage INT = 30000;

-- Loop to insert 10 rows
WHILE @CarID <= 10
BEGIN
    -- Change values each iteration
    SET @Make =
        CASE @CarID % 3
            WHEN 0 THEN 'Toyota'
            WHEN 1 THEN 'Ford'
            ELSE 'Honda'
        END;

    SET @Model =
        CASE @CarID % 3
            WHEN 0 THEN 'Camry'
            WHEN 1 THEN 'Focus'
            ELSE 'Civic'
        END;

    -- Insert row
    INSERT INTO Cars (CarID, Make, Model, Year, Price, Mileage)
    VALUES (@CarID, @Make, @Model, @Year, @Price, @Mileage);

    -- Increment / change variables
    SET @CarID = @CarID + 1;
    SET @Year = @Year + 1;
    SET @Price = @Price + 1200;
    SET @Mileage = @Mileage + 5000;
END;
```

16) Create a table named HikingLocations that has the appropriate structure to remove duplicates when loading the table. Demonstrate that it works. Include sql that defines the structure to remove duplicates in your submission. (this is the same as any other table create except you set the ignore_dup_key option to on).

```sql
CREATE UNIQUE INDEX PK_HikingLocations --creates new index with ignore dupliate key on
ON HikingLocations(HikingLocationsID)
WITH (IGNORE_DUP_KEY = ON);
Go


Insert into HikingLocations --Inserts rows of data
    (HikingLocationsID, name)
values
    (0, 'Mountain'),
    (0, 'Mountain'),
    (0, 'Mountain'),
    (1, 'Waterfall'),
    (2, 'River'),
    (3, 'Overpass'),
    (4, 'Ridge')
```

% No issues found Ln: 1,

Messages

```
Duplicate key was ignored.

(5 rows affected)

Completion time: 2026-04-02T21:17:27.9656521-04:00
```

17) Define a table named HikingLocations2 that has 10 records and 3 of the records are duplicates. Create an SQL routine that identifies the duplicate records by key value. Create a second SQL routine that displays only the duplicate records.

```sql
Insert into HikingLocations2 --Inserts rows of data
    (HikingLocationsID, name)
values
    (0, 'Mountain'), (2, 'Mountain'), (3, 'Mountain'),
    (4, 'Waterfall'), (5, 'River'), (6, 'Overpass'),
    (7, 'Ridge'), (8, 'Trail'), (9, 'Gravel'),
    (10, 'Sidewalk'), (10, 'Concrete'), (10, 'Concrete')

--Counts dup records by key value
Select Name, Count(*) occurrences
FROM HikingLocations2
GROUP BY Name


--Counts dup records by record
Select HikingLocationsID, Name, Count(*) occurences
FROM HikingLocations2
GROUP BY HikingLocationsID, Name
HAVING Count(*) > 1;
```

95 % 11 0

Results Messages

1	Concrete	2
2	Gravel	1
3	Mountain	3
4	Overpass	1
5	Ridge	1
6	River	1
7	Sidewalk	1
8	Trail	1

	HikingLocationsID	Name	occurences
1	10	Concrete	2

18) Create an insert routine for HikingLocations2 that uses variables and performs error checking and error logging.

```sql
Declare @ErrorHold int=0
Insert into HikingLocations2
    (HikingLocationsID, Name)
Values
    (0, 'Mountain')
Set @ErrorHold = @@error --sets the error variable to the error code

If @ErrorHold != 0 --adds error to log
    Insert into ErrorAudit
        (ErrorNbr)
    Values
        (@ErrorHold)
```

5 % 3 0

Messages

```
(1 row affected)

Completion time: 2026-04-02T21:28:32.4043102-04:00
```